

Course: Project Management

Course Code: CENP4201

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Level 4

COLTEC

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Chapter 1

Introduction to Project Management

All of mankind's greatest accomplishments - from building the great pyramids to discovering a cure for polio to putting a man on the moon - began as a project (Larson & Gray, 2011).

There have always been projects and they will always be an integral part of everyday lives of individuals and corporations. Thus projects exist in everything we do; be they from trips to the grocery store to building enterprise level information systems. So for individuals, projects might entail the planning and execution of weekend outings, family-get together, summer trips to Las Vegas or even something as simple as preparing a grocery list that includes allocating grocery budget plus visiting the store to purchase the required items. But your success with the project of going to the grocery store may be judged on whether you were able to purchase everything that you needed within the allocated time frame plus whether you stayed under budget. Failure to meet one or more of these criteria may signify making certain trade-off, such as choosing to buy sausages because you didn't have the budget for beef or buying prepared food that can be prepared in a microwave because you don't have sufficient time to prepare a full four course meal.

Managing trade-off is widespread in an individual's everyday life and it is also crucial in the daily functioning of an organization. Organizations are reliant on the successful execution of projects to survive now and more than ever. Therefore, profitability of an organization is affected by how successful its projects are plus how well they are managed.

Due to the size, complexity and number of IT projects today, Organizations face ever-increasing challenges. This can be demonstrated by the following:

Global U.S. government spending topped \$150 billion in 2007 making U.S. government one of the world's top 3 vertical industries. According to government insights, an IDC company, that's roughly a \$9.1 billion increase. Government insights also forecasts a compound annual growth rate of 4.1% through 2010 (Bozzelli, 2007). Forester Research estimates that global IT spending, purchases of computers, software and services by companies and governments will drop by 3% in 2009 to \$1.66 trillion. This is approximately \$5 billion less than was spent in 2008. Prospects for 2010 were a little bit better

But IT vendors shall brace for the worst through the end of 2009 (Bartels 2008). Lewis Cardin from Forester Research estimates that about 20% of current IT budgets are going into project development (Cook, 2008). One fundamental reason for complexity is that IT changes very rapidly creating a shortage of required IT skill sets because of these differences in IT, organizations are now changing their approach to IT projects making organizational structural changes in order to become smaller and adaptable and better at handling change. So, drifting toward a more supportive model for project management is one way to go

The Project Management Institute (PMI) outlines some benefits for organizations that deploy structured management techniques and they are:

- Improvement in customer satisfaction

- Better cost performance and higher return on investment (ROI)
- Better schedule performance, better allocation of time commitments
- And better utilization of resources, as well as higher productivity
- Increased quality reduces work
- Increased delivering – required features

Consequently, organizations are forced to pay premiums to acquire the required talents through permanent hires, contracting or outsourcing. These resources were required to be onsite (at organization's office to work on project team in the past). But advances in IT have enabled today's IT employee to be able to work from home or work in another region of the country or even in an entirely different country without ever needing to come into office. Geographically dispersed project teams present their own challenges. For the success of a project if organizations have to thrive today, they must embrace project management principles to increase productivity and return on investment (ROI) in order to create a competitive advantage locally as well as globally since the world in which we are living has now become a global village.

Thus our discussion will focus on project management with IT focus since many of the techniques deployed and methods of use are quite different from those in other industries/sectors such as construction, engineering and medicine. We would take a look at these differences plus taking into account the following:

- Traditionally high turnover rates of IT workers
- Level of uniqueness and complexity of each project due to rapid changes in technology
- Difficulty of visualizing software for developers and customer
- Difficulty getting accurate customer requirements
- Rate of change in requirements
- Difficulty testing all possible states of software

So, there is need for constant training to keep team abreast with the technology as shown in Fig 1.1

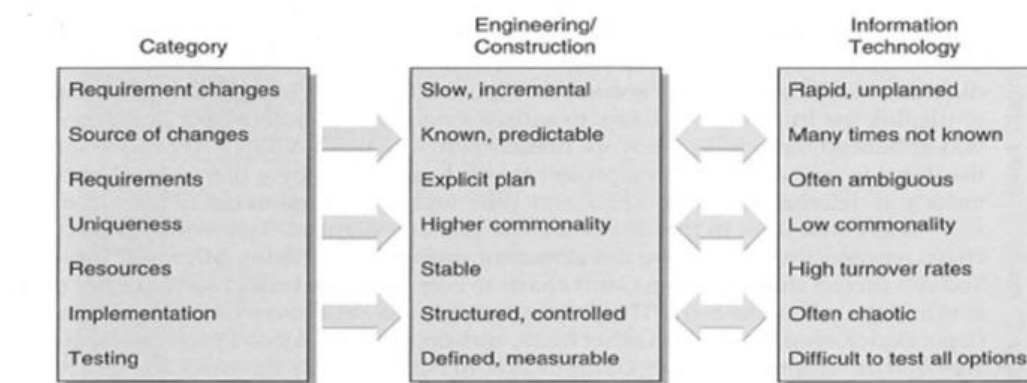


FIGURE 1-1 Project Differences Between Disciplines

Source: Brewer & Dittman, 2010

What is Project/Project Management?

A project is a temporary endeavour undertaken to accomplish a unique purpose (A Guide to the Project Management Body of Knowledge (PMBOK guide))

Project Management consists of the knowledge, skills, methods, technologies and tools used to plan and manage project work. It establishes a sound basis for effective planning, scheduling, resourcing, decision making, management and plan revision (Richman, 2010).

Project management skills assist in completing projects on schedule, within budget and in full accordance with project specifications. Also, they help achieve the organization's other goals such as productivity, quality and cost-effectiveness.

Thus the main objective of project management is to ensure that projects meet agreed goals of time, cost and scope.

History of Project Management

PM has been around since the beginning of time. Noah was a project manager. It took careful planning and execution to build the ark plus gather two of every animal on earth including all the necessary food and water. The Pyramids of Egypt stand today because of the immense projects and project managers.

Although there have been outstanding project managers over the years, PM was not regarded as a formal management concept until operations research in the 1950s and 1960s pioneered methods and specialized tools in managing expensive high-profile aerospace projects such as Polaris and Apollo. As such, NASA and U.S. Department of Defence established project management standards that they expected their contractors to follow. In the middle and late 1960s, business managers began searching for new techniques and organizational structures that would help them adapt rapidly to changing environments. The 1970s and 1980s brought more published data on project management resulting in the development of theories, methods and standards.

The construction industry for example saw prospective benefits of formal project management and began to adopt standards plus develop new techniques. Large-scale initiatives such as quality improvements on reengineering provided data, analysis and problem-solving techniques but no structured discipline to implement them. Therefore, managers turned to project management for direction in implementing and tracking large projects. By 1990s, industries in both profit and nonprofit sectors realized that the size and complexity of their activities were unmanageable without any formal PM processes and tools.

Project Management Today

Modern PM is a leading solution in business operations today. Large and small organizations realize that a structured approach to planning and managing projects is an essential core competency for success.

International organizations such as PM institute (PMI) project and International PM Association (IPMA) promote management by offering professional development programs. PMI offers credentialed certifications such as PMP (Project Management Professional) to those who show competency in the PM field through education and experience and by passing rigorous certification exams. PMI sets standards and accredits degree-granting educational programs in PM. PMI established the first PM Body of Knowledge (PMBOK) in 1987 to document and standardize universally accepted PM information and practices. Current edition, A Guide to the PM Body of Knowledge (PMBOK Guide) is a basic reference for anyone interested in PM. It provides a common lexicon and consistent structure for the field of PM.

PMBOK Guide which is studied and cited by thousands of project managers is frequently updated. So universities offer undergraduate and graduate degree programs in PM and ProjectWorld holds Symposia and seminars throughout the year which are vital opportunities to increase skills, pick up new ideas from other success stories plus network with other professionals.

The Software Crisis

Even though IT is becoming more reliable, faster and expensive, the costs, complexities and risk of IT projects keep on increasing. One consulting firm known as the Standish Group carried out a survey of 365 IT managers in 1995. This widely cited report correctly termed the CHAOS was astonishing e.g. even though U.S. spent over \$250 billion each year on IT application development projects, 31% of these projects were cancelled before their completion, almost 53% were completed but they were over budget and over-scheduled plus didn't meet the original specifications.

Surprised? Strange? Or unbelievable?

Why do you think all this was happening?

On average, cost overrun for a medium-size company survey was 182% of the original estimate while average schedule overrun was 202% i.e. results of this survey are summarized in Table 1.1 below

Table 1.1 Summary of the CHAOS Study Results

<i>Company Size</i>	<i>Average Cost of Development</i>	<i>Average Cost Overruns</i>	<i>Average Schedule Overrun</i>	<i>Original Features and Functions Included</i>	<i>Successful Projects^a</i>	<i>Challenged Projects^b</i>	<i>Impaired Projects^c</i>
Large	\$2,322,000	178%	230%	42%	9%	61.5%	29.5%
Medium	\$1,331,000	182%	202%	65%	16.2%	46.7%	37.1%
Small	\$ 434,000	214%	239%	74%	28%	50.4%	21.6%

^a Completed on-time and on-budget

^b Completed, but over-budget, over schedule, and includes fewer features and functions than originally envisioned

^c Cancelled before completion

SOURCE: Adapted from The Standish Group, *CHAOS* (West Yarmouth, MA: 1995), <http://www.standishgroup.com/visitor/chaos.htm>.

It suggests that a medium-sized originally estimated to cost \$1 million and to take a year to develop actually costs \$1,820,000 took just over 2 years to complete and only included 65% of envisaged features and functions.

Sadly, 48% of IT managers surveyed believed there were more failures at the time than 5 and 10 years earlier

Why Do IT Projects Fail?

The CHAOS report equally provides some amazing insights as to why some projects succeed while others fail. According to survey, user involvement, executive management support plus a clear statement of requirements ranked at the top of the list of factors indispensable for IT project success. On the other hand, non-involvement of users and incomplete requirements seem to be the 2 main factors for projects being challenged or cancelled before completion. Tables 1.1 & 1.2 summarize some of the main findings of the CHAOS report

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Table 1.2 Summary of Factor Rankings for Successful, Challenged, and Impaired Projects

<i>Rank</i>	<i>Factors for Successful Projects</i>	<i>Factors for Challenged Projects</i>	<i>Factors for Impaired Projects</i>
1	User involvement	Lack of user input	Incomplete requirements
2	Executive management support	Incomplete requirements	Lack of user involvement
3	Clear statement of requirements	Changing requirements & specifications	Lack of resources
4	Proper planning	Lack of executive support	Unrealistic expectations
5	Realistic expectations	Technology incompetence	Lack of executive support
6	Smaller project milestones	Lack of resources	Changing requirements specifications
7	Competent staff	Unrealistic expectations	Lack of planning
8	Ownership	Unclear objectives	Didn't need it any longer
9	Clear vision & objectives	Unrealistic time frames	Lack of IT management
10	Hard-working, focused team	New technology	Technology illiteracy

SOURCE: Adapted from The Standish Group, *CHAOS* (West Yarmouth, MA: 1995), <http://www.standishgroup.com/visitor/chaos.htm>.

unpopular project and more and more resources are diverted away from it. The project is barely successful, or a failure.

Larger projects report a success rate of only 9% and seem to be much more risky as compared to medium and smaller projects. Technology, business models and cycle times change too rapidly to develop systems that take much more than a year to complete.

This data also supports the need to break up large projects into smaller, more manageable ones that can be completed in less than a year. User involvement tops the list as the most essential factor in project success. This should not come as a surprise since the client's expertise is required to identify problems and opportunities plus to determine their requirements. Equally active participation by clients keeps them interested and excited about project. Individuals will equally start to take ownership of a project if they feel that they have a stake in the project's success or failure.

Effective communication between techies and non-techies facilitates for a clearer definition of project goals and requirements. Developers and users have more reasonable expectations when working together because they set these expectations together by themselves. So management is more incline to support a popular project while on the other hand, lack of user input or involvement ranks at or near top in factors affecting challenged and impaired projects. One can roughly figure out the series of events

Without close support of key users, the project team could encounter a difficult task understanding goals of the project and defining requirements. This may result in suspicion and hostility may ensue such that there could easily be an "us vs them" situation. Lack of effective communication and a clear direction, changes to project requirements always seem to come into view and both camps might set unrealistic expectations. Thus chaos sets in at this point with the management starting to find fewer reasons to support an unpopular project and more and more resources being diverted away from it. The project is barely successful or failure.

In this regard, the Standish Group has been tracking the management of information technology (IT) projects since 1994. This firm's periodic landmark reports summarize the continued need for improved project management. For over a decade the Standish Reports of management of IT projects showed improvements. In 1994 approximately 16 percent of IT projects were completed on time, on budget; in 2004 the success rate moved up to 29 percent.

However, the CHAOS Summary 2009 report shows a small decrease in the numbers. This survey report shows only 32 percent of IT projects were delivered on time and within budget. However, 44 percent were "challenged," which means they were late, over budget, and/or missed meeting performance requirements. In addition, 24 percent failed, were cancelled, or never used. Jim Crear, Standish Group CIO, notes this is the highest failure rate in over a decade.

Improving the Likelihood of Success

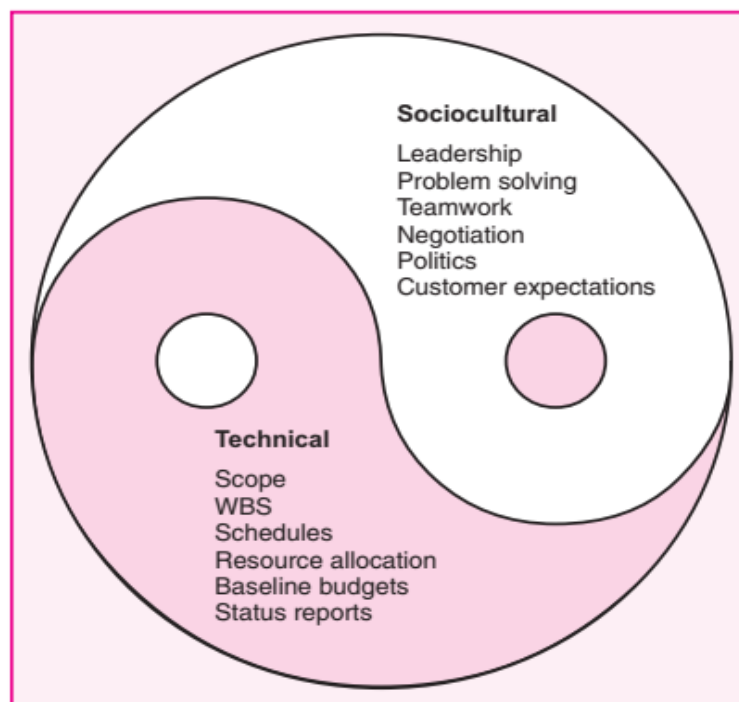
How can we improve chances for IT project success plus avoid repeating past mistakes?

A Socio-Technical Approach

In the past, organizations have attempted the chances of IT project success by focusing on tools, techniques and methodologies of IT development. Solely technical approach, however pays attention to technology. One may end up developing an application which no one asks for or needs. So applications to support e-commerce, supply chain management and integration require that at least equal attention be paid to organizational side. Thus gone are the days of being a good order taker. We are no longer be satisfied with defining a set of user requirements, disappearing for several months and then knocking on the user's door when it's time to deliver the new system.

IT professionals need to understand the business and be actively innovative in deploying the technology in ways that bring value to the organization. As such, clients must become stakeholders in the project. This implies effectively seeking and encouraging their participation, involvement and vision. Figure 1.3 (Larson & Gray. 2011) depicts the various stakeholders involved.

FIGURE 1.3
The Technical and Sociocultural Dimensions of the Project Management Process



Successful application of technology and achievement of project goals must be an equal responsibility of developers and users.

A computerWorld investigation reports that delays in overhauling federal tax system have cost U.S. government approximately \$50 billion a year in uncollected taxes although the Internal Revenue Service (IRS) spent hundreds of millions of dollars in an attempt to modernize its computer systems. Critics believed that a lot of that money has been wasted as a result of mismanagement and primitive development practices.

Government and private groups believe that there are numerous reasons for these problems

- Failure to redesign business processes before beginning system development
- No overall systems architecture or development plan
- Primitive and often “chaotic” software development methodologies
- Failure to manage ISs as investments
- Lack of information security as noted by Marchewka, 2003

A Project Management Approach

One suggestion of CHAOS study was the need for better PM.

But isn't building an IS a project? Have organizations not utilized PM in the past? Or are they not deploying PM now?

Although many organizations have applied PM principles and tools to IT projects, many more put together systems on an ad hoc basis. IT project success or failure depends chiefly on who is or is not part of the team. Deploying PM principles and tools across the whole organization nonetheless should be part of a methodology - step-by-step activities, processes, tools, quality standards, controls and deliverables that are defined for the whole project. Thus, project success doesn't rely solely on the team but more on a set of processes and infrastructure in place. Common set of tools and controls equally provides a universal language across project and ability to compare projects throughout the organization.

In addition, other reasons for PM to support IT projects include:

- Resources: developing or purchasing an IS.
IT projects are capital projects requiring cash plus other organizational resources. Projects must be accurately estimated and costs and schedules must be effectively controlled. Without appropriate tools, techniques, methods and controls in place, projects will divert resources away from other projects plus other areas of the organization. Uncontrolled costs could impact on organization's financial stability.
- Expectations: organizational clients expect IT professionals to deliver quality products and services in a professional manner. Timely status updates and communication plus sound PM practices are needed.
- Competition: internal and external competition has never been greater. Internal IT department's services can easily be outsourced if quality or cost of providing IT services can be bettered outside the organization. Thus competition among consultants today is growing as they compete for business and talents.
- Efficiency and Effectiveness: Peter Drucker, a renowned management guru defined efficiency as doing the thing right and effectiveness as doing the right thing. Several companies note that PM permits for shorter development times, lower costs and higher quality. Simply utilizing PM

tools, nevertheless doesn't guarantee success. PM must become accepted and supported by all levels within the organization plus continued commitment in terms of training, compensation, career path and organizational infrastructure needs to be in place. Support will allow organizations to do the right things and do them right.

A knowledge Management Approach

Socio-technical approach and dedication to PM principles and practices are vital for success. Excellence in IT PM for an individual or an organization takes time and experience. Knowledge management (KM) is a fairly new area.

KM involves a systematic process for acquiring, creating, synthesizing, sharing and using information, insights and experience to transform ideas into business value (Marchewka, 2003). Even though several organizations have KM initiatives in progress today, investing on KM systems is expected to increase

So what about learning from experience?

Experience is a great teacher but these experiences and knowledge gained are often patchy across the organization. Chances are that if you encounter what appears to be a unique problem situation, someone else within your organization might have perhaps dealt with that problem or one very similar. Wouldn't it be great to inquire from that person what they did and even what was the outcome? And would they do it again in the same manner? Sadly, this individual might be of the other side of the globe or just down the hall – of which you may not even have an idea!

Knowledge and experience in form of lessons can be documented plus made obtainable through technologies accessible today. They comprise the WWW or a local version of the web called intranets. Lessons learned that document both reasons for success and failure embody valuable asserts if maintained and deployed correctly. Usually, anyone who gains experience is regarded as being more mature. In like manner, an organization that learns from its experiences could be in its processes by taking those lessons learned plus producing best practices i.e. basically executing activities in most efficient and effective manner.

In case of IT projects, managing knowledge in the form of lessons learned can assist an organization to develop best practices that permit every project teams within organization to do the right things plus do them right